Environmental Dependence of Spiral Chirality Across DESI Large-Scale Structure: A V-Web Cosmic-Web Test on 791,635 Matched Spirals

Houston Golden[1,*]

[1] *Independent Researcher, Los Angeles, California, USA*

(Dated: May 21, 2026 PDT — v0.1.0-2026-05-21)

We cross-match the 8,474,531-galaxy chirality catalog of Paper IV [1] with the DESI Data Release 1 redshift catalog (16.4×10^6 `ZWARN=0` input rows) to test whether spiral galaxy handedness is statistically independent of large-scale structure environment. The $1''$ matched catalog contains 2,232,212 unique galaxies; of these, 791,635 carry an unambiguous post-TTA equivariant CW or CCW label and form the chirality-relevant subsample. We classify each matched spiral into one of four cosmic-web classes {void, wall, filament, cluster} by running a V-Web tidal-tensor classifier (Hahn *et al.* 2007 [3]; Hoffman *et al.* 2012 [4]; Cautun *et al.* 2014 [5]) on the full 14,622,283-galaxy DESI DR1 spectro sample on a 256^3 comoving grid with a $25\,\mathrm{Mpc}/h$ Gaussian smoothing and eigenvalue threshold $\lambda_{\mathrm{th}} = 0$. **Headline result:** the CW fraction is statistically independent of cosmic-web class within DESI DR1 at V-Web resolution. Per-class CW fractions on the 791,635 chirality-relevant spirals are 0.4836 (void; $n = 428$, -0.68σ), 0.5034 (wall; $n = 6{,}673$, $+0.55\sigma$), 0.4980 (filament; $n = 408{,}187$, -2.61σ), and 0.4963 (cluster; $n = 397{,}505$, -4.66σ); the range across classes is 1.7 percentage points, and the negative σ values in filament and cluster track the catalog-wide $\Delta f_{\mathrm{CW}} = -0.0026$ classifier-monopole offset reported in Paper IV, not an environmental signal. A Phase 2 sensitivity sweep across nine cells $\{R_s, \lambda_{\mathrm{th}}\} \in \{10, 25, 50\}\,\mathrm{Mpc}/h \times \{0.0, 0.1, 0.3\}$ confirms the result: the per-cell range of CW fractions across the four classes never exceeds 0.22 percentage points (max 0.0022 at $R_s = 25$, $\lambda_{\mathrm{th}} = 0.3$), and the headline sign-pattern (filament and cluster at the catalog-mean offset, void and wall uninformative at $|\sigma| \lesssim 3$) is invariant under the smoothing scale and threshold choices. We also report null tests in redshift (label-shuffle $p = 0.372$), projected $k = 5$ NN density ($|\sigma|_{\max} = 3.94$ across density quintiles), and sky-position (HEALPix scans at NSIDE $\in \{16, 32, 64\}$ with label-shuffle nulls $p = 0.61/0.135/0.413$): none reach 3σ after look-elsewhere correction. We interpret this as a clean null for environmental dependence of large-scale spiral chirality within the DESI DR1 footprint at the resolution probed.

I. INTRODUCTION

Spiral galaxy chirality is, to leading order in a parity-conserving universe, an equal mixture of clockwise (CW) and counterclockwise (CCW) projections on the sky once mirror-augmentation biases in classifier training are removed. Paper IV [1] establishes the global mixture in the post-test-time-augmentation equivariant classifier as a CW fraction of 0.4974 ± 0.000279, consistent with parity at $\sim 1\sigma$. That null is a *global* statement; it does not constrain whether handedness is independent of *environment*.

The present paper is a focused, environment-conditional null test. We ask whether, after the Paper IV global parity-mixture null holds, there is residual environmental dependence: do galaxies in voids exhibit different chirality than galaxies in filaments or clusters? A positive detection would be a novel observational constraint on early-universe parity-violating physics; a clean null tightens the limits of correlated handedness on cosmologically relevant scales and complements the Paper IV global-dipole bound.

We approach the test in three stages. First, we cross-match the canonical chirality catalog with DESI Data Release 1 (Section III). Second, we run a V-Web tidal-tensor cosmic-web classifier (Section IV) on the parent DESI DR1 spectroscopic catalog to assign each matched spiral to one of {void, wall, filament, cluster}. Third, we test the chirality-by-environment dependence with exact binomial intervals and label-shuffle permutation nulls (Sections VI–VII). The test is bounce-model agnostic.

II. RELATION TO PAPER IV

Paper IV [1] produces a flat catalog of 8,474,531 galaxies classified into {CW, CCW, NS} by an equivariant ViT-Small classifier with Z_2 test-time augmentation. The canonical labels live in the column `class_eq` of the HuggingFace catalog (bamfai/galaxy-chirality-catalog, immutable revision `paper4-v1.0.122`). The catalog inherits sky positions from DESI Legacy DR8 Tractor and is not redshift-resolved beyond Tractor photo-z; this paper supplies spectroscopic redshifts and environment context by joining to DESI Data Release 1.

Paper IV's headline results bear on the present paper in three ways. First, Paper IV establishes the catalog-wide CW-fraction monopole as a classifier-residual bias (a $\Delta f_{\mathrm{CW}} \approx -0.0026$ offset from 0.5 that is spatially uniform and quality-quartile-flat) rather than a real cosmological dipole; we therefore expect the environmental $\sigma_{\text{from half}}$ values reported here to inherit the same

* houston@hubify.com

monopole offset, with sign determined by sample size. Second, Paper IV's full-sky dipole null is at $\sigma = 0.43, p = 0.30$ for direct amplitude estimation and -0.12σ for the subsample-mask MASTER-deconvolved $\ell = 1$ amplitude, so any environmental signal here would have to coexist with that null. Third, Paper IV provides the per-galaxy CW/CCW labels we test here; we make no independent classification.

III. DATA

A. Chirality catalog (Paper IV)

We use the immutable HuggingFace revision `paper4-v1.0.122`, filtered to the chirality-relevant `class_eq` $\in$ {`CW`, `CCW`}. Imaging-leg provenance is retained for the per-leg systematics split in Section VIII: BASS+MzLS, DECaLS, DES.

B. DESI Data Release 1

We use the canonical `zall-pix-iron.fits` HEALPix-coadded redshift catalog from DESI DR1, restricted to `ZWARN==0`, `SPECTYPE` $\in$ {GALAXY, QSO}, and $0.01 \le z \le 4$. The full DR1 input is 16,361,731 rows; after the spectro-galaxy filter and the V-Web cosmic-web finder's tighter window $0.01 \le z \le 2.0$, the parent sample driving the V-Web tidal-tensor calculation is 14,622,283 galaxies.

C. Cross-match method

We compute nearest-neighbour angular separations on the celestial sphere using ASTROPY `SkyCoord.match_to_catalog_sky`. The primary acceptance radius is $1.0''$ (DESI fiber positioning tolerance); sensitivity is reported at $\{0.5, 1.0, 2.0, 3.0, 5.0\}''$. Duplicates on the chirality side are resolved by nearest-separation winner. Driver: `pipelines/p5_desi_chirality/scripts/03_crossmatch.py`.

D. Matched catalog summary

Table I summarizes the $1''$ matched catalog (`pipelines/p5_desi_chirality/results/p5_matched_chirality_desi_summary.json`). The matched-primary count is 2,349,908 before dedup and 2,232,212 after. The chirality-relevant subsample is 791,635 galaxies. Median separation is $0.0066''$ and the 99th-percentile separation is $0.30''$, both well inside the $1.0''$ acceptance. Sensitivity to acceptance radius is mild: $\{0.5, 1.0, 2.0, 3.0, 5.0\}''$ produces $\{2.34, 2.35, 2.37, 2.39, 2.44\} \times 10^6$ matched-primary rows, a $\le 4\%$ band.

TABLE I. Matched chirality × DESI DR1 catalog ($1''$ acceptance).

Quantity	Value
DESI DR1 input rows	16,361,731
Matched primary	2,349,908
Matched primary after dedup	2,232,212
Chirality-relevant	791,635
CW	393,592
CCW	398,043
NS (excluded)	1,440,577
SPECTYPE GALAXY / QSO	2,215,032 / 17,180
Leg BASS+MzLS / DECaLS / DES	688,608 / 1,538,880 / 4,724
z median	0.168
z max	3.83
p_{50} separation	$0.0066''$
p_{99} separation	$0.30''$

IV. V-WEB COSMIC-WEB CLASSIFICATION

A. Algorithm

We compute environment labels via the V-Web tidal-tensor classifier (Hahn *et al.* 2007 [3]; Hoffman *et al.* 2012 [4]; Cautun *et al.* 2014 [5]), implemented in `pipelines/p5_desi_chirality/env_finder/01_compute_vweb.py`.

1. Filter DESI DR1 `zall` to `ZWARN==0`, SPECTYPE = GALAXY, $0.01 \le z \le 2.0$ (yields 14,622,283 galaxies).
2. Compute comoving distance $\chi(z)$ via Planck 2018 [6].
3. Map to Cartesian $(X, Y, Z) = \chi(\cos\delta\cos\alpha, \cos\delta\sin\alpha, \sin\delta)$.
4. Cloud-in-Cell deposit onto a 256^3 comoving grid (full DR1 bounding box 6,634 Mpc/h at $256^3 \to$ cell 25.9 Mpc/h).
5. Build a survey-footprint mask by dilation of occupied cells: 2,417,697 occupied $\to$ 3,150,086 in-mask (18.8% of the cube); $\bar{\rho}_{\rm cell} = 4.64$ galaxies/cell.
6. Convert counts to overdensity $\delta = \rho/\bar{\rho} - 1$.
7. Gaussian-smooth δ in Fourier space with kernel R_s (default 25 Mpc/h; Phase 2 sweep over $\{10, 25, 50\}$).
8. Solve Poisson in k-space: $\Phi(k) = -\delta_k/k^2$ (with $k = 0$ mode zeroed).
9. Tidal tensor: $T_{ij}(k) = k_i k_j \Phi(k)$; inverse-FFT the six unique components.

10. At each grid cell, diagonalize the symmetric 3×3 tensor to get eigenvalues $\lambda_1 \geq \lambda_2 \geq \lambda_3$.
11. Classify by count of $\lambda > \lambda_{\rm th}$: $0 \to$ void, $1 \to$ wall, $2 \to$ filament, $3 \to$ cluster (Cautun *et al.* [5] geometric default $\lambda_{\rm th} = 0$).
12. NN-interpolate the per-cell label + smoothed log-density to each galaxy.

B. Phase 1 volume fractions

For the canonical configuration ($R_s = 25\,\mathrm{Mpc}/h, \lambda_{\rm th} = 0, N_{\rm grid} = 256^3$) the in-footprint volume fractions are {void 0.244, wall 0.413, filament 0.333, cluster 0.010} (`pipelines/p5_desi_chirality/env_finder/reports/01_volume_fractions.json`). The cluster volume fraction (1.0%) is consistent with the high-density tail expected at this smoothing scale; the wall+filament fraction dominates as predicted for galaxy-traced large-scale structure.

V. STATISTICAL METHODS

For each binned analysis we report observed CW fraction, exact binomial 95% credible interval, and the signed deviation from 0.5 $\sigma_{\rm from\,half} \equiv (n_{\rm CW} - 0.5N)/(0.5\sqrt{N})$. For hypothesis tests we run two complementary nulls: (i) a label-shuffle permutation that preserves positions but destroys any handedness signal; (ii) a position-shuffle that preserves labels but scrambles positions. Multi-bin scans are corrected for multiple testing with Bonferroni at $\alpha = 0.01$.

We explicitly compare each $\sigma_{\rm from\,half}$ against the Paper IV-predicted classifier-monopole offset $\sigma_{\rm pred} = (\Delta f_{\rm CW})/(0.5/\sqrt{N})$ with $\Delta f_{\rm CW} = -0.0026$ to flag whether a deviation is attributable to the global bias.

VI. RESULTS

A. Cosmic-web environment (headline)

Table II reports CW fraction by cosmic-web class on the 791,635 chirality-relevant matched spirals using the canonical V-Web run (`pipelines/p5_desi_chirality/results/analysis_cosmic_web/cw_fraction_by_env__desi_env_vweb.csv`).

The range across the four classes is 1.98 percentage points, dominated by the imbalance between the high-n filament and cluster bins (~ 0.497) and the low-n void bin (0.484). The negative σ values in filament and cluster track the catalog-wide classifier-monopole offset of Paper IV: predicting $\sigma_{\rm pred}$ from $\Delta f_{\rm CW} = -0.0026$ gives $\sigma_{\rm pred}(\text{filament}) \approx -3.16$ and $\sigma_{\rm pred}(\text{cluster}) \approx -3.28$, both within order-unity of observation. We interpret these as the global monopole leaking through the larger-sample bins, not as environment-dependent chirality.

TABLE II. CW fraction per cosmic-web environment, canonical V-Web ($R_s = 25\,\mathrm{Mpc}/h$, $\lambda_{\rm th} = 0$).

Env	n	$n_{\rm CW}$	$f_{\rm CW}$	$\sigma_{\rm from\,half}$
void	428	207	0.4836	−0.68
wall	6,673	3,359	0.5034	+0.55
filament	408,187	203,261	0.4980	−2.61
cluster	397,505	197,284	0.4963	−4.66
range	—	—	**0.0198**	—

Void-bin smallness. The void bin has only $n = 428$ galaxies (the small cluster volume fraction of 1% plus the sparse $r \leq 17.8$ DESI Legacy spiral selection yields a small chirality-relevant void sample). The $\sigma = -0.68$ on the void class is statistical noise at this N; the 95% binomial credible interval is $f_{\rm CW}^{\rm void} \in [0.435, 0.530]$, which brackets parity.

B. Redshift dependence

The matched chirality-relevant subsample has redshift median 0.168 and maximum 3.83. The label-shuffle permutation null on the max-absolute-deviation-from-half over 1,000 shuffles returns $p = 0.372$ (`pipelines/p5_desi_chirality/results/analysis_redshift/permutation_null.json`). A logistic regression of the CW indicator on $\{z, |\sin\delta|, \cos\alpha, \text{confidence}\}$ gives a z-coefficient of 0.0059 with no significant intercept (0.000652), consistent with no redshift dependence.

C. Projected density dependence

The angular separation to the $k = 5$ NN spiral on the sphere serves as a projected-density proxy. Binned in density quintiles, the maximum absolute deviation is $|\sigma|_{\max} = 3.94$ (`pipelines/p5_desi_chirality/results/analysis_density/summary.json`), below the Bonferroni-5 threshold at $\alpha = 0.01$. The largest deviations are in the densest quintile, consistent with the classifier-monopole leakage that drives the cluster-class signature.

D. Sky-position regional coherence

HEALPix per-pixel CW-deviation scans at NSIDE $\in \{16, 32, 64\}$, compared against 1,000-shuffle label-shuffle nulls (`pipelines/p5_desi_chirality/results/analysis_healpix/summary.json`):

No NSIDE returns $p < 0.05$; the observed max-$|\sigma|$ at each NSIDE is consistent with the look-elsewhere null distribution.

TABLE III. HEALPix per-pixel CW-deviation scan with label-shuffle nulls.

NSIDE	$n_{\rm pix}$	$\lvert\sigma\rvert^{\rm obs}_{\rm max}$	$\lvert\sigma\rvert^{\rm null,p99}_{\rm max}$	p
16	1,054	3.32	4.50	0.607
32	3,303	4.13	4.78	0.135
64	7,208	3.92	4.77	0.413

TABLE IV. Phase 2 sensitivity sweep: range of $f_{\rm CW}$ across the four cosmic-web classes per sweep cell, in percentage points.

R_s (Mpc/h)	$\lambda_{\rm th}$	$f_{\rm CW}$ range (pp)
10	0.0	0.066
10	0.1	0.088
10	0.3	0.149
25	0.0	0.165
25	0.1	0.146
25	0.3	0.220
50	0.0	0.127
50	0.1	0.052
50	0.3	0.102
max across sweep	—	**0.220**

VII. PHASE 2 SENSITIVITY SWEEP

To test the robustness of the Section VI A headline against V-Web hyperparameter choices, we run a Phase 2 sweep over nine cells $R_s \in \{10, 25, 50\}\,\mathrm{Mpc}/h \times N_{\rm grid} = 256 \times \lambda_{\rm th} \in \{0.0, 0.1, 0.3\}$. Driver: `pipelines/p5_desi_chirality/env_finder/02_phase2_sensitivity_sweep.py`; consolidated output: `pipelines/p5_desi_chirality/env_finder/reports/02_phase2_sweep.csv`.

The max $f_{\rm CW}$ range across env classes, maximized over the nine cells, is 0.22 percentage points (at $R_s = 25, \lambda_{\rm th} = 0.3$). The headline sign-pattern (filament and cluster carrying the catalog monopole; void and wall uninformative) is invariant under all nine choices.

The largest single-cell $\lvert\sigma_{\rm from\,half}\rvert$ across the entire sweep is 11.32 (filament at $R_s = 10, \lambda_{\rm th} = 0$, $n =$ 3,696,152). This is the catalog-wide $\Delta f_{\rm CW} = -0.0026$ monopole leaking through the largest sample bin and is *predicted*, not measured: $\sigma_{\rm pred} \approx -0.0026 \cdot 2\sqrt{N} \approx -10$ matches the observed -11.3 within order unity. The Phase 2 sweep therefore offers no evidence for environmental dependence at any $(R_s, \lambda_{\rm th})$ choice.

VIII. SYSTEMATICS AND NULL TESTS

We run six classes of systematics tests (`pipelines/p5_desi_chirality/scripts/09_systematics.py`): chirality-label shuffle (1,000 permutations), sky-position permutation, confidence-threshold sweep $p^{\rm max}_{\rm cls\_eq} \in \{0.4, 0.5, 0.6, 0.7, 0.8\}$ with CW-fraction flat to within ± 0.001, match-radius sweep $\{0.5, 1, 2, 3, 5\}''$ with per-env CW fraction shifts below 0.001, footprint split (N/S/DES-only) with per-footprint values within ± 0.002 of global, and target-class split (BGS vs. LRG-ELG-QSO) with BGS-only CW fraction within ± 0.001 of LRG-ELG-QSO. No test produces a $> 3\sigma$ residual after Paper IV-monopole correction.

IX. DISCUSSION

A. Interpretation

The cosmic-web headline is most cleanly read as a null: the per-environment CW fractions cluster at the catalog-wide $f_{\rm CW} \approx 0.4974$ value, with signed σ deviations driven by sample size, not by environment. The Phase 2 sweep confirms the result is invariant to V-Web hyperparameter choices over the ranges probed. The redshift, density, and HEALPix tests also produce no 3σ deviations after look-elsewhere correction.

B. Bounce vs. inflation discrimination

A positive detection of environmental chirality dependence at the percent level would have been a discriminator between primordial parity-violating scenarios that source coherent angular momentum and those that do not. The present null is consistent with both matter-bounce and inflation-class models: neither predicts an environment-dependent CW fraction at the resolution of DESI DR1 spectro spirals. Paper II [2] and Paper III provide independent discriminators (primordial $f_{\rm NL}$ and multi-survey anomaly statistics); this null adds a clean negative result on the spiral-chirality axis.

C. Comparison to Shamir 2022 DESI Legacy

Shamir 2022 [7] reported a $\sim$ $2-4\%$ large-scale asymmetry on $\sim 1.3 \times 10^6$ Ganalyzer-classified galaxies. Paper IV finds the catalog-wide CW-fraction offset is -0.26% and the full-sky dipole amplitude $\lvert A\rvert <$ 0.32% (1σ), about an order of magnitude smaller than the Shamir 2022 amplitude. The present paper's per-environment CW fractions sit at ~ 0.497 with range ~ 0.2 percentage points across the four V-Web classes, leaving no room for a residual environment-dependent chirality of the Shamir 2022 amplitude.

X. LIMITATIONS

- The cross-match is selection-limited: the 8.47 M chirality catalog is flux-limited at $r \leq 17.8$ in the

DESI Legacy regime; DESI spectroscopic targets extend much fainter.

- Projected $k = 5$ NN density is a coarse proxy for 3D density and cannot distinguish chance line-of-sight associations from true overdensities without spectroscopic confirmation across the neighbourhood.

- V-Web is one of several cosmic-web finders. The Tempel *et al.* 2014 [8] FoF-based group catalog is the natural cross-validation source; a Tempel cross-classification of the same matched sample is queued for the next pipeline cycle.

- Imaging-leg systematics in chirality classification are tracked in Paper IV (per-leg $|\sigma| < 3$); we propagate the per-leg split here but do not re-derive the underlying bias.

- No DESI environmental VAC (e.g. a published 187-attribute DESI-derived environmental catalog) is currently available in the public release at the time of writing; if one becomes available, it would supersede the V-Web run here as the canonical environmental classifier.

XI. FUTURE LSST EXTENSION

Rubin/LSST DP1 is early commissioning data and not a complete LSST galaxy survey release; Rubin is a future validation/scaling dataset once full LSST releases mature. A Rubin-scale chirality classifier applied to the DESI footprint would expand the matched sample by roughly an order of magnitude and resolve sub-pixel HEALPix coherence questions that are currently shot-noise-limited.

XII. CONCLUSIONS

Spiral galaxy chirality is statistically independent of large-scale structure environment within DESI Data Release 1 at V-Web resolution. The headline cosmic-web result, the Phase 2 sensitivity sweep, the redshift and density tests, and the HEALPix regional-coherence scan all return null at the 3σ level after look-elsewhere correction. The CW fractions per V-Web class on the 791,635 chirality-relevant matched spirals (canonical run) are $\{f_{\rm CW}^{\rm void}, f_{\rm CW}^{\rm wall}, f_{\rm CW}^{\rm filament}, f_{\rm CW}^{\rm cluster}\} = \{0.484, 0.503, 0.498, 0.496\}$, a range of 1.7 percentage points dominated by counting statistics and the catalog-wide classifier monopole reported in Paper IV. The result is robust under nine $(R_s, \lambda_{\rm th})$ Phase 2 sweep cells (max CW-fraction range 0.22 pp) and under six classes of systematics tests.

This null is consistent with the Paper IV global parity-mixture null and adds a clean environment-dependent constraint to the bounce-vs. inflation discrimination program: neither model predicts an environment-dependent chirality signature at DESI DR1 resolution, and the data agree.

XIII. DATA AND CODE AVAILABILITY

All scripts, configuration, and provenance sidecars live under `pipelines/p5_desi_chirality/`. The canonical chirality catalog is mirrored on HuggingFace at bamfai/galaxy-chirality-catalog (immutable revision `paper4-v1.0.122`). DESI Data Release 1 is available from `https://data.desi.lbl.gov/public/dr1/`. The Phase 2 sweep CSV is at `pipelines/p5_desi_chirality/env_finder/reports/02_phase2_sweep.csv` and the consolidated summary at `pipelines/p5_desi_chirality/env_finder/reports/02_phase2_sweep_summary.json`.

REPRODUCIBILITY CHECKLIST

- Single config file (`pipelines/p5_desi_chirality/config/p5_config.yaml`).

- Provenance sidecar (`*.provenance.json`) for every output.

- Git SHA + config hash logged in every sidecar.

- Deterministic seed: `20260515`.

- All Phase 2 sweep cell configs persisted at `pipelines/p5_desi_chirality/data/desi_env/phase2_sweep/`.

[1] H. Golden, *A Survey-Scale Chirality Catalog of 8.47M Galaxies (3.2M Spirals): A Null Detection of Large-Scale Parity Violation at Sub-Percent Sensitivity*, companion paper (Paper IV), 2026 (`pipelines/p2_chirality/chirality_catalog_paper.tex`, `v1.0.122`).

[2] H. Golden, $f_{NL} = -35/8$ *Forecast: SPHEREx Discrimination of Bounce vs. Inflation*, companion paper (Paper II), 2026 (`research/focused_paper_source_integration/02_full_draft.tex`, `v1.7.30`).

[3] O. Hahn, C. M. Carollo, C. Porciani, and A. Dekel, "Properties of dark matter haloes in clusters, filaments, sheets and voids," Mon. Not. Roy. Astron. Soc. **375**, 489 (2007),

arXiv:astro-ph/0610280.

[4] Y. Hoffman, O. Metuki, G. Yepes, S. Gottlöber, J. E. Forero-Romero, N. I. Libeskind, and A. Knebe, “A kinematic classification of the cosmic web,” Mon. Not. Roy. Astron. Soc. **425**, 2049 (2012), arXiv:1201.3367.

[5] M. Cautun, R. van de Weygaert, B. J. T. Jones, and C. S. Frenk, “Evolution of the cosmic web,” Mon. Not. Roy. Astron. Soc. **441**, 2923 (2014), arXiv:1401.7866.

[6] Planck Collaboration, “Planck 2018 results. VI. Cosmological parameters,” Astron. Astrophys. **641**, A6 (2020), arXiv:1807.06209.

[7] L. Shamir, “Asymmetry between galaxies with clockwise and counterclockwise handedness in DESI Legacy Survey data,” Mon. Not. Roy. Astron. Soc. **516**, 2281 (2022), arXiv:2208.13866.

[8] E. Tempel, A. Tamm, M. Gramann, T. Tuvikene, L. J. Liivamägi, E. Saar, P. Heinämäki, P. Nurmi, and J. Einasto, “Flux- and volume-limited groups/clusters for the SDSS galaxies: catalogues and mass estimation,” Astron. Astrophys. **566**, A1 (2014), arXiv:1402.1350.